

## Performance Testing

|               |              |
|---------------|--------------|
| Date          | 7 July 2026  |
| Team ID       | XXXX         |
| Project Name  | Smart Lender |
| Maximum Marks | 5 Marks      |

## Performance Testing

### Step 1: Testing Overview

|                     |                                        |
|---------------------|----------------------------------------|
| Field               | Details                                |
| Testing Tool Used   | Postman                                |
| Type of Testing     | Functional Performance Testing         |
| Target Module / API | Loan Eligibility Prediction (/predict) |
| Test Environment    | Local Flask Server / Render Deployment |
| Test Date           | 07 July 2026                           |

### Step 2: Test Scenarios

| S.No | Test Scenario / Description                      | No. of Virtual Users | Duration (sec) | Expected Outcome                                             |
|------|--------------------------------------------------|----------------------|----------------|--------------------------------------------------------------|
| 1    | Submit valid applicant information               | 1                    | 30             | Prediction generated successfully                            |
| 2    | Submit multiple prediction requests sequentially | 10                   | 60             | Stable response without errors                               |
| 3    | Submit invalid or incomplete input               | 5                    | 30             | Input validation prevents incorrect requests                 |
| 4    | Continuous prediction requests                   | 20                   | 120            | Application remains responsive with acceptable response time |

### Step 3: Performance Test Results

| S.No | Metric                  | Target Value | Actual Value | Status | Remarks                  |
|------|-------------------------|--------------|--------------|--------|--------------------------|
| 1    | Response Time (Average) | < 2 seconds  | 0.82 sec     | Pass   | Fast prediction response |

|   |                         |                   |                 |      |                               |
|---|-------------------------|-------------------|-----------------|------|-------------------------------|
| 2 | Response Time (Maximum) | < 5 seconds       | 1.67 sec        | Pass | Within acceptable limit       |
| 3 | Throughput              | > 10 Requests/sec | 14 Requests/sec | Pass | Stable request handling       |
| 4 | Error Rate              | < 1%              | 0%              | Pass | No failed prediction requests |
| 5 | CPU Utilization         | < 80%             | 12%             | Pass | Efficient CPU usage           |
| 6 | Memory Utilization      | < 80%             | 35%             | Pass | Stable memory consumption     |

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#### Step 4: Observations & Analysis

- The application responded consistently to prediction requests without noticeable delays.
- Input validation prevented invalid data from reaching the prediction engine.
- The Flask backend successfully handled multiple prediction requests with stable response times.
- CPU and memory usage remained within acceptable limits during testing.
- No prediction failures or application crashes were observed.
- The deployed application demonstrated reliable performance for the expected user workload.
- The application is suitable for small to medium-scale usage and can be further optimized for higher traffic if required.